

MA1521 L1 Week 05 Wed 20260909

In each of the following, a function f , a point P , and a vector $\mathbf{A}$ are given. Calculate the gradient of f at the point, and the directional derivative $\left. \frac{df}{ds} \right|_{\mathbf{u}}$ at the point, in the direction $\mathbf{u}$ of the given vector $\mathbf{A}$.

a) $x^3 + 2y^3$; $(1, 1)$, $\mathbf{i} - \mathbf{j}$

b) $w = \frac{xy}{z}$; $(2, -1, 1)$, $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$

c) $z = x \sin y + y \cos x$; $(0, \pi/2)$, $-3\mathbf{i} + 4\mathbf{j}$

d) $f(u, v, w) = (u + 2v + 3w)^2$; $(1, -1, 1)$, $-2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$

In the following, $x = \cos t$, $y = \sin t$, and $w = f(x(t), y(t))$ where $f(x, y)$ is an unknown function

a) Suppose $\nabla w = 2\mathbf{i} + 3\mathbf{j}$ at $P : (1, 0)$; Find the value of $\frac{dw}{dt}$ at $t = 0$.

b) Suppose $\nabla w = y\mathbf{i} + x\mathbf{j}$; Find $\frac{dw}{dt}$ and tell for what t -values it is zero.

In the following, $x = t$, $y = t^2$, $z = t^3$, and $f(x, y, z)$ is an unknown function

a) Suppose $\nabla f = (1, -1, 2)$ at $(1, 1, 1)$. Find $\frac{df}{dt}$ at $t = 1$.

b) Suppose $\nabla f = (3x^2y, x^3 + z, y)$; Find $\frac{df}{dt}$

For the following functions, each with a given point P ,

- (i) find the maximum and minimum values of $\left. \frac{df}{ds} \right|_{\mathbf{u}}$, as $\mathbf{u}$ varies;
- (ii) tell for which directions the maximum and minimum occur;
- (iii) find the direction(s) $\mathbf{u}$ for which $\left. \frac{df}{ds} \right|_{\mathbf{u}} = 0$.

a) $w = \ln(3x - 4y), \quad (1, 1)$

b) $w = xy + yz + xz, \quad (1, -1, 2)$

Suppose $\left. \frac{dw}{ds} \right|_{\mathbf{u}} = 2$, $\left. \frac{dw}{ds} \right|_{\mathbf{v}} = 1$ at P , where $\mathbf{u} = \frac{\mathbf{i} + \mathbf{j}}{\sqrt{2}}$, $\mathbf{v} = \frac{\mathbf{i} - \mathbf{j}}{\sqrt{2}}$. Find $(\nabla w)_P$.

Let $w = f(x, y)$, and assume that $\nabla w = 2\mathbf{i} + 3\mathbf{j}$ at $(0, 1)$. If $x = u^2 - v^2$, $y = uv$, find $\frac{\partial w}{\partial u}$, $\frac{\partial w}{\partial v}$ at $u = 1$, $v = 1$.

Let $w = f(x, y)$, and suppose we change from rectangular to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

a) Show that $(w_x)^2 + (w_y)^2 = (w_r)^2 + \frac{1}{r^2} (w_\theta)^2$.

b) Suppose $\nabla w = 2\mathbf{i} - \mathbf{j}$ at the point $x = 1, y = 1$.

Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial \theta}$ when $r = \sqrt{2}, \theta = \pi/4$

The Jacobian matrix for the change of variables $x = x(u, v)$, $y = y(u, v)$ is defined to be $J = \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$. Let $\nabla f(x, y)$ be represented as the row vector $\langle f_x \ f_y \rangle$.

Show that

$$\nabla f(x(u, v), y(u, v)) = \nabla f(x, y) \cdot J \quad (\text{matrix multiplication}).$$